Assignment questions on **C# File Handling**

🧑‍🏫 Assignment 1: Employee Log Management System

🎯 Scenario

You are building a logging system for a company where employee login/logout details must be recorded daily.

📌 Requirements

1. Create a console application.
2. When an employee logs in:
    - Append data to a file employee_log.txt
    - Format:
    - EmployeeId | Name | LoginTime | LogoutTime
3. Allow user to:
    - Add login entry
    - Update logout time for an employee
4. Use:
    - StreamWriter (Append mode)
    - StreamReader to read/update records
5. Ensure:
    - File is created automatically if it doesn't exist
    - Proper exception handling (try-catch-finally)

💡 Expected Learning

- File creation and append operations
- Reading and updating file content
- Handling structured text data

🧑‍🏫 Assignment 2: Student Report Card Generator

🎯 Scenario

A school wants to generate and store student report cards in files.

📌 Requirements

1. Accept student details:
    - Name, Roll Number, Marks (3 subjects)
2. Calculate:
    - Total
    - Average
    - Grade (A/B/C/Fail)
3. Save each student report as a separate file:
4. File Name: RollNumber.txt
5. File content format:
6. Student Name:
7. Roll Number:
8. Marks:
9. Total:
10. Average:
11. Grade:
12. Add feature to:
    - Read and display a student report by entering Roll Number

💡 Constraints

- Use File.WriteAllText() / File.ReadAllText()
- Validate input data

💡 Expected Learning

- Writing structured data into files

- File naming conventions
- Reading entire file content

👩‍🏫 Assignment 3: Mini Notepad (Text Editor)

🎯 Scenario

Create a simple Notepad-like application in console.

📌 Requirements

1. Menu-driven program:
2. 1. Create New File
3. 2. Write to File
4. 3. Read File
5. 4. Append Text
6. 5. Delete File
7. Functionalities:
    - Create a .txt file
    - Write multiple lines (use loop)
    - Append new content without deleting old content
    - Read and display file content
    - Delete file if required
8. Use:
    - FileStream, StreamWriter, StreamReader
9. Handle:
    - File not found errors
    - Unauthorized access exceptions